



Proposal Report: Project ASCEND

IROC-U 2026 Challenge

Team Name: RotorHawk

Team ID: 10427

January 2026

Sri Krishna College of Engineering and Technology, Coimbatore - 641008

Contents

1	Description of ASCEND	5
1.1	Overview of ASCEND	5
1.2	Functional Block Schematic	5
1.3	Comparative Analysis of Rotorcraft Configurations	6
1.4	Justification for the Selected Quadcopter Approach	6
2	System Architecture	7
2.1	Sub-system Enumeration and Interfaces:	7
2.2	Operational Logic and Interdependencies:	9
3	Identification of Components and Specifications	10
3.1	Mechanical Components	10
3.2	Electrical and Power Systems	10
3.3	Sensors and Communication	10
3.4	Algorithms and Software Platform	11
4	ASCEND Realization Plan	12
4.1	Hardware Realization Plan	12
4.2	Software Implementation Plan	12
5	Testing Plan	14
5.1	Sub-System Level Testing	14
5.1.1	Propulsion and Power Stress Test	14
5.1.2	VIO and SLAM Accuracy Verification	14
5.1.3	Landing Pad Mechanical Coupling Test	14
5.2	System Level Testing	14
5.2.1	Autonomous Boustrophedon Navigation Test	15
5.2.2	Full Mission Profile (End-to-End)	15
6	Brief of System Specification	16

6.1	Performance Capability Overview	16
7	Overview of the Emergency Response System	17
7.1	Identification of Emergency Situations	17
7.2	Proposed Emergency Response System	17
7.2.1	Failsafe working principles	17
7.3	Rationale for the Proposed System	18
8	Project Management	19
8.1	Team Responsibilities and System Breakdown	19
8.2	Schedule and Strategy for Management	19
8.3	Cost Estimation	19
9	Novelty in the Overall Proposal	20
9.1	Asynchronous Batch AI Processing	20
9.2	Mechanical Innovations	20

List of Figures

1	Functional Block Diagram of the ASCEND System	5
2	Flight Control Sub-system	7
3	Perception Sub-system	7
4	Navigation and SLAM Sub-system	8
5	Recognition Sub-system	8
6	Docking and Energy Sub-system	9

List of Tables

1	Comparison of Rotorcraft Configurations.	6
2	Drone System Interface Specifications	9
3	Summary Comparison of Critical Components	11
4	ASCEND Hardware Realization and Procurement Plan	12

5	Summary of Testing Matrix and Verification Criteria	15
6	ASCEND System Specification Summary	16
7	Emergency Situation and Response Matrix	18
8	Team Responsibility Matrix and System Breakdown	19
9	Preliminary Project Budget and Cost Estimation.	19

1 Description of ASCEND

This section outlines the core purpose of the **ASCEND** project, details its functional architecture, and justifies the selection of a quadcopter platform for the IROC-U 2026 challenge.

1.1 Overview of ASCEND

ASCEND is an autonomous quadcopter engineered for high-fidelity mapping and feature recognition in GNSS-restricted, Martian-simulated environments. The system employs a dual-processor architecture: a **Pixhawk 2 Black Cube** for low-latency flight stabilization and a **Raspberry Pi 5 (8 GB RAM) with 13 TOPS AI HAT+** for VIO, RTAB-Map SLAM, and post-landing batch image processing. The mission objective is to execute a boustrophedon survey to generate an arena orthomosaic and log coordinates of seeded features.

1.2 Functional Block Schematic

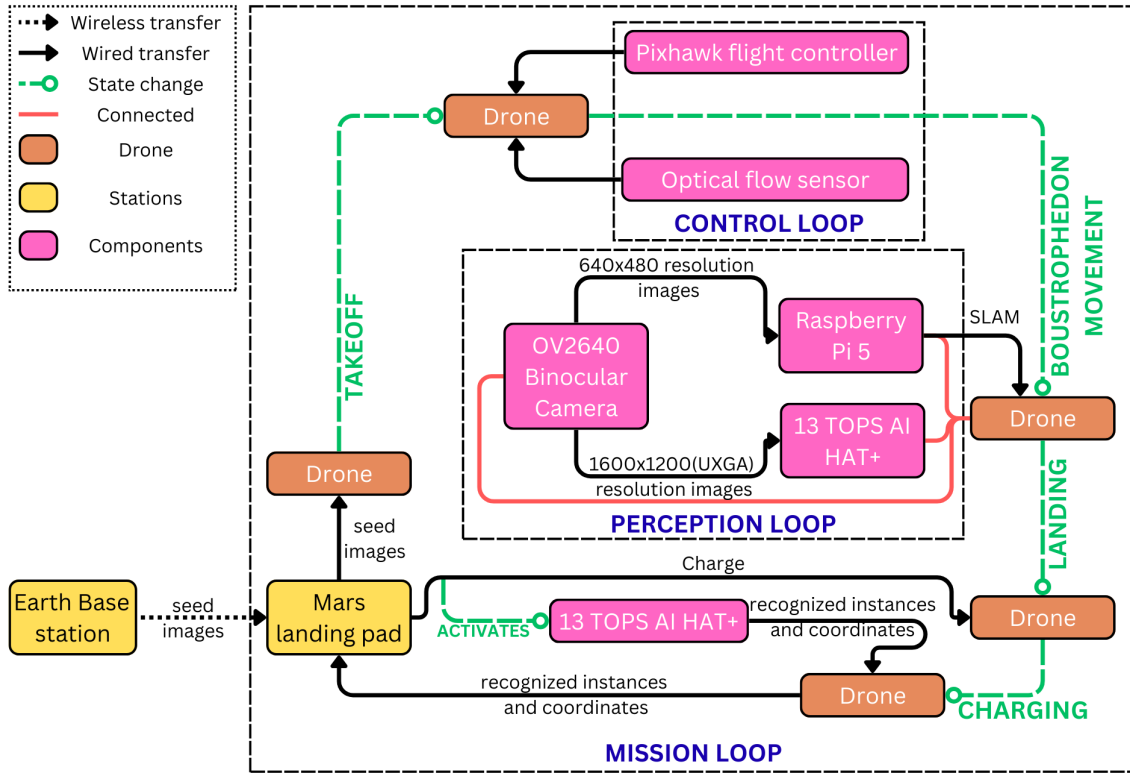


Figure 1: Functional Block Diagram of the ASCEND System

The interdependencies of the ASCEND subsystems are illustrated in the above figure. The system operates through three primary data loops:

- **Control Loop:** Pixhawk fuses IMU, optical flow, and barometric data for attitude and altitude stability.

- **Perception Loop:** Stereo OV2640 camera transmits dual-stream feeds (640x480 for VIO; 1600x1200 for archival) to the RPi 5 for real-time localization and mapping.
- **Mission/Data Loop:** Post-landing, an ESP32-managed rotating ring initiates pogo-pin contact, triggering 125MB data transfer after completion of batch AI classification.

1.3 Comparative Analysis of Rotorcraft Configurations

Potential configurations were evaluated against the 10x10m arena constraints and <2kg mass limit.

Configuration	Advantages	Limitations for ASCEND
Fixed-Wing	High energy efficiency.	No VTOL/Hover; large turn radius; unsuitable for 10x10m arena.
Hexacopter	High redundancy and payload.	Exceeds 2kg mass limit; high power consumption.
Helicopter	High hover efficiency.	High vibration noise; mechanical complexity; safety risks.
Quadcopter	Agile; mechanically simple; optimal weight-to-thrust.	Ideal balance for VTOL and high-compute payloads.

Table 1: Comparison of Rotorcraft Configurations.

1.4 Justification for the Selected Quadcopter Approach

The quadcopter platform was selected for its synergy with autonomous mission requirements:

- **VTOL & Precision Hover:** Essential for boustrophedon survey consistency and autonomous dome-docking.
- **Maneuverability:** Symmetrical X-frame enables rapid heading changes and stable lateral movement for 60% side-overlap mapping.
- **Weight Efficiency:** Utilizing four 2212 motors maintains a 2:1 thrust-to-weight ratio at approximately 1.6 kg, leaving a 400g margin for structural reinforcements.
- **Maturity:** Robust open-source support for MAVLink/ROS2 integration between Pixhawk and Raspberry Pi 5.

In conclusion, the quadcopter provides the most stable and computationally efficient platform within the 2026 challenge constraints.

2 System Architecture

The ASCEND architecture is a modular, high-redundancy system designed for autonomous exploration in GPS-denied Martian environments. The architecture is divided into three primary functional loops—Control, Perception, and Mission—interfacing to ensure seamless transitions from survey to recovery.

2.1 Sub-system Enumeration and Interfaces:

The architecture comprises five core sub-systems:

Flight Control Sub-system (Control Loop):

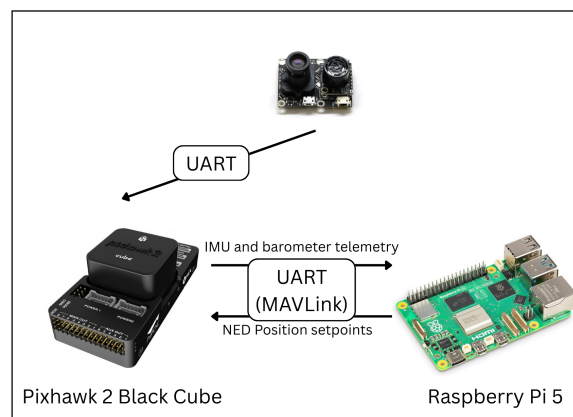


Figure 2: Flight Control Sub-system

Centered on the Pixhawk Flight Controller, this system manages low-level flight dynamics. It interfaces via MAVLink with the companion computer and receives real-time velocity data from the Optical Flow Sensor.

Perception Sub-system (Perception Loop):

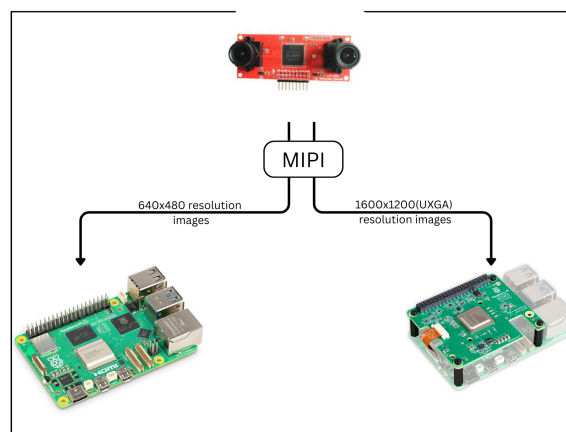


Figure 3: Perception Sub-system

Utilizing a Stereo OV2640 Binocular Camera, this sub-system performs dual-stream processing. It provides a 640x480 feed for visual odometry and a 1600x1200 (UXGA) feed for archival feature logging.

Navigation and SLAM Sub-system:

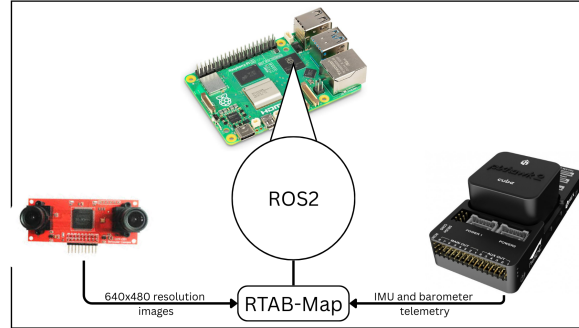


Figure 4: Navigation and SLAM Sub-system

Hosted on the Raspberry Pi 5, this system runs the RTAB-Map library to generate a global map with loop closure. It interfaces with the Pixhawk to provide high-level position setpoints derived from Visual-Inertial Odometry (VIO).

Recognition Sub-system:

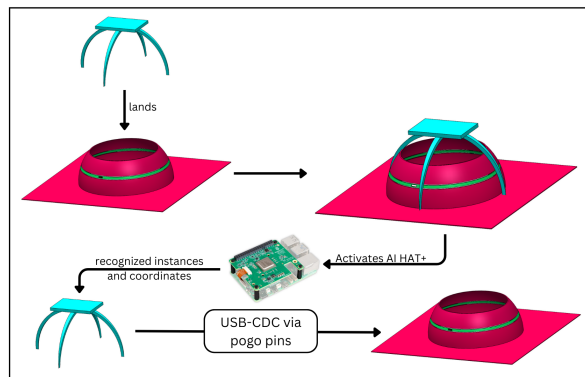


Figure 5: Recognition Sub-system

Driven by the 13 TOPS AI HAT+, this system performs batch-processing of archival images post-landing to classify features and generate accurate coordinate logs.

Docking and Energy Sub-system (Mission Loop):

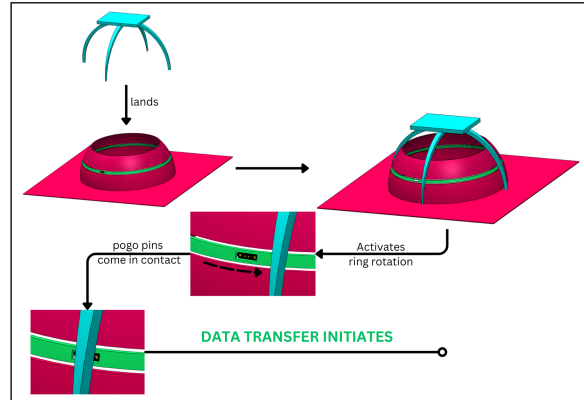


Figure 6: Docking and Energy Sub-system

Managed by an ESP32, this sub-system controls the home station's rotating ring and pogo-pin deployment. It interfaces with the drone via physical contact for high-speed USB data transfer and 4S battery charging.

2.2 Operational Logic and Interdependencies:

The system follows a hierarchical logic flow. During the Boustrophedon Movement, the Perception Loop provides the SLAM data required for navigation. Upon reaching the end of the arena or a critical battery threshold, the Mission Loop initiates the Landing phase. The final Charging state triggers the 13 TOPS AI HAT+ to activate, transitioning the system from real-time navigation to batch feature quantification and data sync via the pogo-pin interface.

Table 2: Drone System Interface Specifications

From Sub-system	To Sub-system	Interface Type	Data/Function
Perception	Navigation	CSI / MIPI	Stereo Video Feed (VGA/UXGA)
Navigation	Flight Control	UART (MAVLink)	NED Position Setpoints
Flight Control	Navigation	UART (MAVLink)	IMU and Barometer Telemetry
Home Station	Recognition	USB / Pogo-Pin	Coordinate & Image Transfer
Home Station	Drone Battery	Concentric Pads	4S Charging Current

3 Identification of Components and Specifications

This section details the hardware and software components selected for Project ASCEND. The architecture is optimized to meet the strict 2.0 kg mass limit while providing sufficient computational power for GPS-denied autonomous navigation.

3.1 Mechanical Components

The mechanical design prioritizes structural rigidity for stable VIO and a specialized docking interface for autonomous charging.

- **Structure (Frame):** A custom 3D-printed/Carbon-fiber hybrid X-frame (300mm wheel-base). It is designed to accommodate the dome-docking legs and the active gimbal mount.
- **Propulsion (Motors & Propellers):** Four **DYS D2826-13 1000KV** brushless motors paired with **10x4.5 propellers**. This setup provides a high thrust-to-weight ratio to compensate for the 10Ah battery mass while maintaining agility for survey maneuvers.
- **Actuators (Active Gimbal):** A 2-axis servo-based gimbal (utilizing SG90 servos) to maintain the **Nadir (downward-facing)** camera orientation regardless of the UAV's flight pitch or roll.

3.2 Electrical and Power Systems

The electrical system is designed to support high-current recharging and sustained computational loads.

- **UAV Power:** A **3S 11.1V 5000mAh 35C LiPo** battery. This capacity was chosen based on a power-to-weight justification to ensure at least 10 minutes of flight time, providing a significant safety margin for the assumed 10x10m arena survey.
- **Home Station Power:** A **4S 14.8V 20,000mAh** battery pack. The higher voltage allows for efficient DC-to-DC charging of the drone's 3S pack via the pogo-pin interface.
- **Companion Computer:** **Raspberry Pi 5 (8GB)** paired with a **13 TOPS AI HAT+**. This provides the necessary RAM and NPU acceleration for RTAB-Map SLAM and post-landing batch classification.
- **Flight Controller:** **Pixhawk 2 Black Cube**, selected for its robust MAVLink integration and low-latency IMU data processing.

3.3 Sensors and Communication

- **Optical Sensor:** **OV2640 Binocular Camera (Stereo)**. It provides a dual-stream MIPI-CSI feed: 640x480 for real-time VIO and 1600x1200 (UXGA) for high-resolution feature archival.

- **Inertial & Velocity Sensors:** An integrated **Optical Flow Sensor** and the Pixhawk’s internal IMU/Barometer. These provide the primary odometry source for horizontal and vertical stabilization.
- **Wireless Communication:** **ESP32-S2** module with a **5dBi external antenna**. It utilizes the WebSocket protocol over 2.4GHz for low-latency bi-directional telemetry and native USB support for docking data transfer.

3.4 Algorithms and Software Platform

- **Software Platform:** The system runs on **Ubuntu 22.04** with **ROS2 Humble**.
- **Mapping & Localization:** **RTAB-Map (Real-Time Appearance-Based Mapping)** provides V-SLAM with loop closure.
- **Classification:** A YOLO-based bounding box approach running asynchronously on the **13 TOPS NPU** during the charging state.

Table 3: Summary Comparison of Critical Components

Component	Selection	Justification
Compute	RPi 5 + AI HAT	Highest TOPS-per-watt for batch feature recognition.
Communication	ESP32-S2	Integrated USB OTG support for high-speed docking sync.
Battery	10Ah 3S LiPo	Optimized for 20-minute mission profile and high-current recovery.
SLAM	RTAB-Map	Robust loop-closure and stereo camera support in ROS2.

4 ASCEND Realization Plan

Upon identifying the necessary components, the team has formulated a comprehensive realization plan to transition from baseline design to a functional system. Hardware components are classified by their source of realization—whether directly procured, 3D printed, or custom-fabricated—to ensure a streamlined assembly process.

4.1 Hardware Realization Plan

The following table categorizes all identified hardware for the micro-UAV and the Base-station, outlining procurement sources and specific realization strategies.

Table 4: ASCEND Hardware Realization and Procurement Plan

No.	Hardware Details	Procurement Source	Specifications / Realization Plan	Qty
1	Motors	Market	DYS D2826-13 1000KV; High-torque brushless	4
2	Frame Structure	3D Printed / Fabrication	Carbon Fiber-reinforced PETG; Custom X-frame	1
3	Propellers	Market	10x4.5 Inch; Balanced composite material	4
4	Flight Controller	Market	Pixhawk 6C; MAVLink compatible	1
5	Companion Computer	Market	Raspberry Pi 5 (8GB)	1
6	AI Accelerator	Market	13 TOPS AI HAT+	1
7	Optical Sensor	Market	OV2640 Binocular Stereo Camera	1
8	UAV Battery	Market	3S 11.1V 5000mAh 35C LiPo	1
9	Base Station Battery	Market	4S 14.8V 20,000mAh Pack	1
10	Active Gimbal	3D Printed / Market	SG90 Servos with 3D printed Nadir mount	1
11	Docking Station	Fabrication / 3D Print	Dome-shaped structure with rotating ring	1
12	Data Interface	Market / Fabrication	ESP32-S2 with custom Pogo-pin housing	1

4.2 Software Implementation Plan

The software implementation strategy follows a modular ROS2-based approach to ensure robust communication between the various sub-systems identified in the mission flow.

- **OS and Middleware:** The primary operating system is Ubuntu 22.04 LTS, utilizing **ROS2 Humble** as the communication middleware for node management and message passing.
- **SLAM and Navigation Layer:** Implementation of the **RTAB-Map** library to facilitate stereo V-SLAM. This involves configuring nodes to fuse 640x480 video streams with IMU and Optical Flow data for real-time localization.
- **Control Interface:** Developing a MAVROS-based bridge to translate ROS2 high-level navigation commands into MAVLink packets for the Pixhawk flight controller.

- **Vision and Batch Processing:** Leveraging the Hailo-8L drivers on the **13 TOPS AI HAT+** to implement a YOLO-based classification pipeline. This is designed for asynchronous batch processing of UXGA images stored during flight.
- **Docking Logic:** Implementing an ESP32-based state machine for the Home Station. This logic handles limit-switch debouncing, servo actuation for pogo-pin deployment, and the subsequent USB data synchronization.

5 Testing Plan

The testing plan for ASCEND is designed to verify that each sub-system and the integrated platform perform reliably under Martian-simulated conditions within the 10m x 10m arena. This section details the hierarchical testing approach, moving from component validation to full mission profile verification.

5.1 Sub-System Level Testing

Sub-system tests ensure that individual hardware and software modules meet their performance specifications before integration.

5.1.1 Propulsion and Power Stress Test

- **Test Plan:** Secure the UAV to a static test bench and operate the DYS D2826-13 motors at 50% (hover) and 100% (max) throttle for 10 minutes to simulate a full battery cycle. Monitor thermal data from the ESCs and voltage sag from the 5000mAh 3S battery.
- **Expected Outcome:** The propulsion system must maintain constant thrust without thermal throttling, and the battery must not drop below 3.3V per cell within the 10-minute window.

5.1.2 VIO and SLAM Accuracy Verification

- **Test Plan:** Move the Raspberry Pi 5 and OV2640 stereo camera manually along a pre-measured 5m path. Record the estimated position from RTAB-Map and compare it against physical measurements.
- **Expected Outcome:** The translation error should remain below 5% of the total distance traveled, and loop closure must trigger successfully upon returning to the start point.

5.1.3 Landing Pad Mechanical Coupling Test

- **Test Plan:** Place the drone at various arbitrary orientations on the dome-shaped landing pad. Trigger the ESP32-governed rotating ring to detect the drone legs and deploy the pogo-pins.
- **Expected Outcome:** The dome geometry must center the drone via gravity, the limit switches must trigger within 360 degrees of rotation, and the pogo-pins must achieve a stable electrical connection.

5.2 System Level Testing

Integrated tests verify the interdependencies and mission logic across all sub-systems.

5.2.1 Autonomous Boustrophedon Navigation Test

- **Test Plan:** Execute a full 10m x 10m lawnmower survey pattern at an altitude of 4m. Use the 640x480 VIO stream for real-time drift correction and the internal barometer for altitude hold.
- **Expected Outcome:** The UAV must maintain a path deviation of less than 0.5m and achieve the required 60% image overlap for subsequent orthomosaic generation.

5.2.2 Full Mission Profile (End-to-End)

- **Test Plan:** Perform a complete mission sequence: Takeoff → Search Phase → Landing on Dome → Charging/Data Transfer → Batch Recognition on AI HAT+.
- **Expected Outcome:** Successful autonomous landing and charging, followed by the generation of a feature coordinate log with at least 90% classification accuracy on the 13 TOPS NPU.

Table 5: Summary of Testing Matrix and Verification Criteria

Test Category	Verification Metric	Success Threshold
Flight Dynamics	Hover Stability	$\pm 10\text{cm}$ deviation (Optical Flow)
Perception	Mapping Resolution	2.89 mm/pixel GSD at 4m
Communication	WebSocket Latency	< 50ms round-trip
Docking	Contact Reliability	100% connectivity in 10 trials
AI Inference	Batch Processing Time	< 120s for 125MB of UXGA images

6 Brief of System Specification

The following table summarizes the technical specifications and capabilities of the ASCEND system. This specification provides a general overview of the rotorcraft’s size, power requirements, and specialized features integrated for the IRoC-U 2026 challenge.

Table 6: ASCEND System Specification Summary

Sl. No.	Description	Specification
1	Overall mass	1.78 kg (Take-off weight)
2	Overall dimensions	300 mm (Diagonal Wheelbase)
3	Power requirements	11.1V (UAV - 3S LiPo) / 14.8V (Base Station - 4S Li-ion)
4	Flight time per charge	12–15 minutes (Optimized for 10m x 10m arena mapping)
5	Number of propellers	4 (Quad-rotor configuration)
6	Propeller size	10 x 4.5 Inch (1045)
7	4 Key Features	Dual-stream Perception: 640x480 (VIO) / 1600x1200 (UXGA)
		Autonomous Dome-Docking with self-centering geometry
		Asynchronous Batch AI Processing via 13 TOPS NPU
		Servo-actuated Pogo-pin USB Data Transfer Interface
8	Others	ROS2 Humble Middleware; RTAB-Map V-SLAM with Loop Closure

6.1 Performance Capability Overview

Based on the specifications outlined above, the ASCEND system is capable of executing a full arena survey with a 60% image overlap at a 4-meter altitude. The shift to a 5,000 mAh battery ensures the total mass remains below the 2.0 kg limit while providing a 3x safety margin over the estimated 5-minute mission duration. The propulsion system, utilizing DYS D2826-13 motors, provides a thrust-to-weight ratio of approximately 2.2:1, ensuring high agility during the boustrophedon search phase.

7 Overview of the Emergency Response System

The ASCEND system incorporates a multi-layered emergency response framework to mitigate risks during autonomous flight, mapping, and docking within the 10m x 10m arena. This system identifies potential hardware and software failures and implements predefined failsafe protocols to ensure the safety of the rotorcraft and the surrounding environment.

7.1 Identification of Emergency Situations

We have identified the following critical emergency situations anticipated during the task execution:

- **Loss of SLAM / Tracking Failure:** Occurs when the OV2640 camera cannot identify enough visual features for RTAB-Map to maintain a position lock.
- **Communication Heartbeat Loss:** Disruption of the WebSocket connection between the Raspberry Pi 5 and the Pixhawk or the Ground Station.
- **Critical Battery Depletion:** Voltage drop below 3.5V per cell in the 3S 5,000 mAh flight pack.
- **Docking Collision/Misalignment:** Failure of the dome-centering mechanism or pogo-pin obstruction during the rotating ring sequence.

7.2 Proposed Emergency Response System

The proposed system utilizes a priority-based state machine to address these situations.

7.2.1 Failsafe working principles

1. **Degraded Navigation Mode (V-SLAM Loss):** In the event of a SLAM failure, the system immediately reverts to pure **Optical Flow and Barometric feedback**. The Pixhawk enters a "Position Hold" state, halting the boustrophedon pattern until a visual relocalization is achieved.
2. **Automated Land-on-Spot:** If communication heartbeat loss exceeds 2.5 seconds or battery levels reach the critical 15% threshold, the Pixhawk executes a controlled vertical descent to land at the current NED coordinates to prevent unpredictable drift.
3. **Virtual Geofencing:** Utilizing the yellow boundary detection from the perception loop, the drone is software-restricted from exiting the 10m x 10m arena. If the boundary is breached, the drone is forced into a "Brake" state.
4. **Mechanical Data Link Cut-off:** The ESP32-S2 monitors the current draw of the pogo-pin servos. If a stall is detected (indicating an obstruction), the servos retract immediately to prevent damage to the rotating ring.

7.3 Rationale for the Proposed System

Alternative emergency systems, such as a simple "Return to Home" (RTH) upon any failure, were considered. However, in a GPS-denied environment where the "Home" coordinate depends on the same SLAM system that may have failed, RTH can be hazardous. The proposed "Land-on-Spot" and "Degraded Navigation" approach was chosen because it prioritizes immediate stabilization and minimizes the risk of collisions with arena boundaries or personnel.

Table 7: Emergency Situation and Response Matrix

Situation	Primary Response	Redundancy
SLAM Tracking Lost	Switch to Optical Flow Hold	Search for Loop Closure
Low Battery (20%)	Initiate Landing Sequence	Visual Warning to Base
Comms Heartbeat Lost	Hover for 5s then Land	Auto-Retract Pogo Pins
Propeller Damage	Immediate Motor Cut-off	Physical Guard Protection

8 Project Management

The ASCEND project utilizes a structured management strategy to ensure all subsystems—Control, Perception, and Mission loops—are developed, tested, and integrated within the competition timeline. Responsibilities are delineated based on the specialized engineering backgrounds of the team members from Sri Krishna College of Engineering and Technology.

8.1 Team Responsibilities and System Breakdown

The project is divided into core technical domains as illustrated in the responsibility matrix below.

Table 8: Team Responsibility Matrix and System Breakdown

No.	Task / Domain	Main Responsibility	Deadline	Secondary
1	SLAM Algorithms	Abdullah (Lead)	TBD	Dinesh D.
2	Telemetry & Power	Monish R.	TBD	Yogeshwaran T.
3	Altitude & Stability	Navith Sanjay N.	TBD	Dinesh M.
4	Structure Design	Aswin S.	TBD	Abdullah
5	UI & Imaging Sensors	Sanjiv B.	TBD	Hareshkumar Y.
6	Hardware Procurement	Dharshaan P. V.	TBD	Aswin S

8.2 Schedule and Strategy for Management

The team follows an iterative development schedule to meet the high-fidelity mapping requirements of the Martian arena.

- **Phase 1: Baseline Design:** Finalization of the dual-processor architecture and selection of the 5,000 mAh power system.
- **Phase 2: Sub-system Fabrication:** Simultaneous 3D printing of the dome-landing station and development of the WebSocket communication layer.
- **Phase 3: Integration and Testing:** Full-system validation of the Boustrophedon movement and batch AI classification.

8.3 Cost Estimation

A preliminary budget has been established to stay within the financial constraints of a student-led project.

Category	Key Components	Estimated Cost (INR)
Propulsion	D2826 Motors, 1045 Props, ESCs	8,500
Computation	RPi 5, AI HAT+, Pixhawk 6C	18,000
Power System	5000mAh 3S UAV, 20Ah 4S Base	7,500
Sensors & Comms	OV2640, ESP32-S2, Optical Flow	4,500
Miscellaneous	3D Printing, Frame, Pogo Pins	3,000
Total Estimated Budget		41,500

Table 9: Preliminary Project Budget and Cost Estimation.

9 Novelty in the Overall Proposal

Project ASCEND introduces original engineering solutions to overcome the computational and mechanical constraints of Martian exploration drones.

9.1 Asynchronous Batch AI Processing

In contrast to traditional real-time classification, ASCEND utilizes an **asynchronous processing workflow**.

- **Computational Optimization:** By deferring UXGA feature recognition until the drone is docked, 100% of the 13 TOPS NPU power is dedicated to classification accuracy without risking in-flight thermal throttling or navigational lag.
- **Data Fidelity:** This architecture enables the use of high-resolution 1600x1200 images for classification, ensuring superior quantification compared to the low-resolution streams used in real-time VIO.

9.2 Mechanical Innovations

The system features two primary mechanical advancements designed for high-dust, GPS-denied environments:

- **Servo-Actuated Pogo Pin Interface:** To prevent friction-corrosion, pogo pins remain retracted within the docking ring during rotation. They are deployed by a servo only after a limit-switch confirms alignment, ensuring a high-integrity USB 2.0 connection.
- **Gravity-Aided Dome Self-Centering:** The 45-degree dome geometry passively corrects landing misalignments up to 15cm. This eliminates the need for power-intensive precision-landing algorithms, allowing the UAV to prioritize energy conservation.